Network Analysis

AN INTRODUCTION FOR HUMANISTS

Dr Katarzyna Anna Kapitan

11 March 2026

Recap

- Hubs
- Degree Centrality
- Closeness Centrality
- Betweenness Centrality
- Robustness

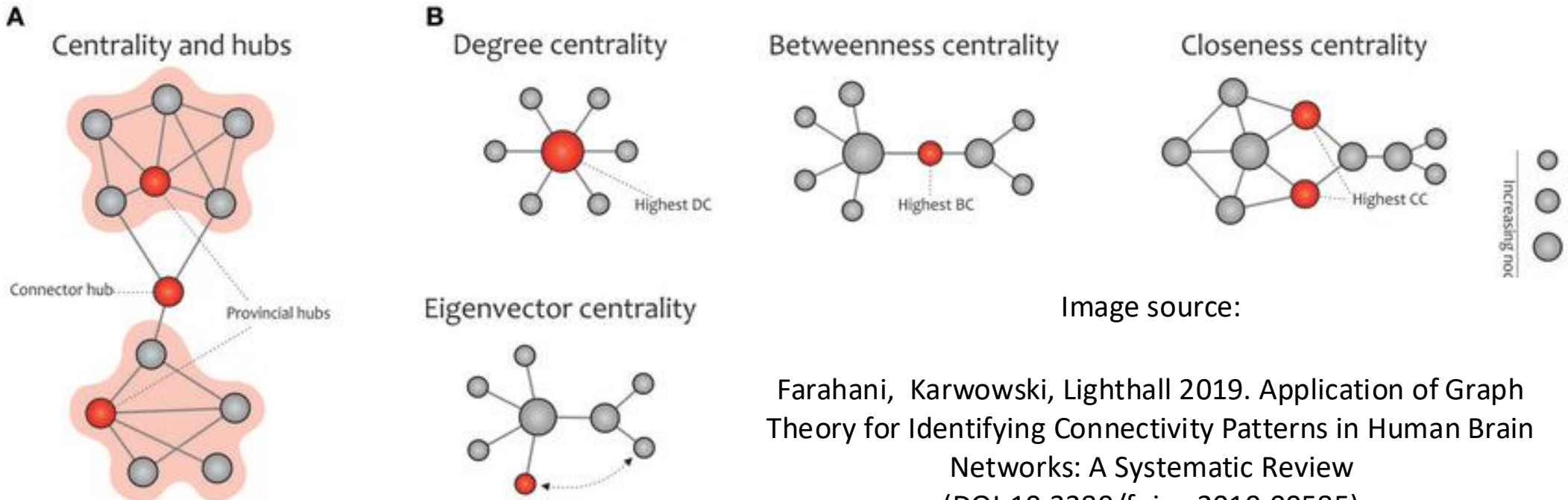


Image source:

Farahani, Karwowski, Lighthall 2019. Application of Graph Theory for Identifying Connectivity Patterns in Human Brain Networks: A Systematic Review (DOI:10.3389/fnins.2019.00585)

Today: Measuring Importance and Influence 2.0

Eigenvector Centrality

Katz Centrality

PageRank

Measuring Importance and Influence

Degree centrality: assignes a point to a node (vertex) for every neighbour in the network.

‘But not all neighbours are equivalent. In many circumstances a vertex's importance in a network is increased by having connections to other vertices that are *themselves important.’* (Newman 2010, *Networks: An Introductio*n)

Politics: friends with the president of a country = friends with an administrative assistant of a party?
YES/NO?

How to measure that?

Eigenvector Centrality

Eigenvector Centrality
(**eigencentrality** or **prestige score**) is a common measures used to quantify node's importance and influence in the network.

'Instead of awarding vertices just one point for each neighbour, eigenvector centrality gives each vertex a score proportional to the sum of the scores of its neighbours'
(Newman 2010, *Networks: An Introduction*)

Eigenvector Centrality

The eigenvector centrality of node *i* (vertex *i*) is proportional to the sum of the centralities of *i's neighbours*.

- ‘The eigenvector centrality [has] the nice property that it can be large either because **a vertex has many neighbours** or because it **has important neighbours** (**or both**). An individual in a social network, for instance, can be important, by this measure, because he or she knows lots of people (even though those people may not be important themselves) or knows a few people in high places.’ (Newman 2010, *Networks: An Introduction*)

- **A high eigenvector score means that a node is connected to many nodes who themselves have high scores.**

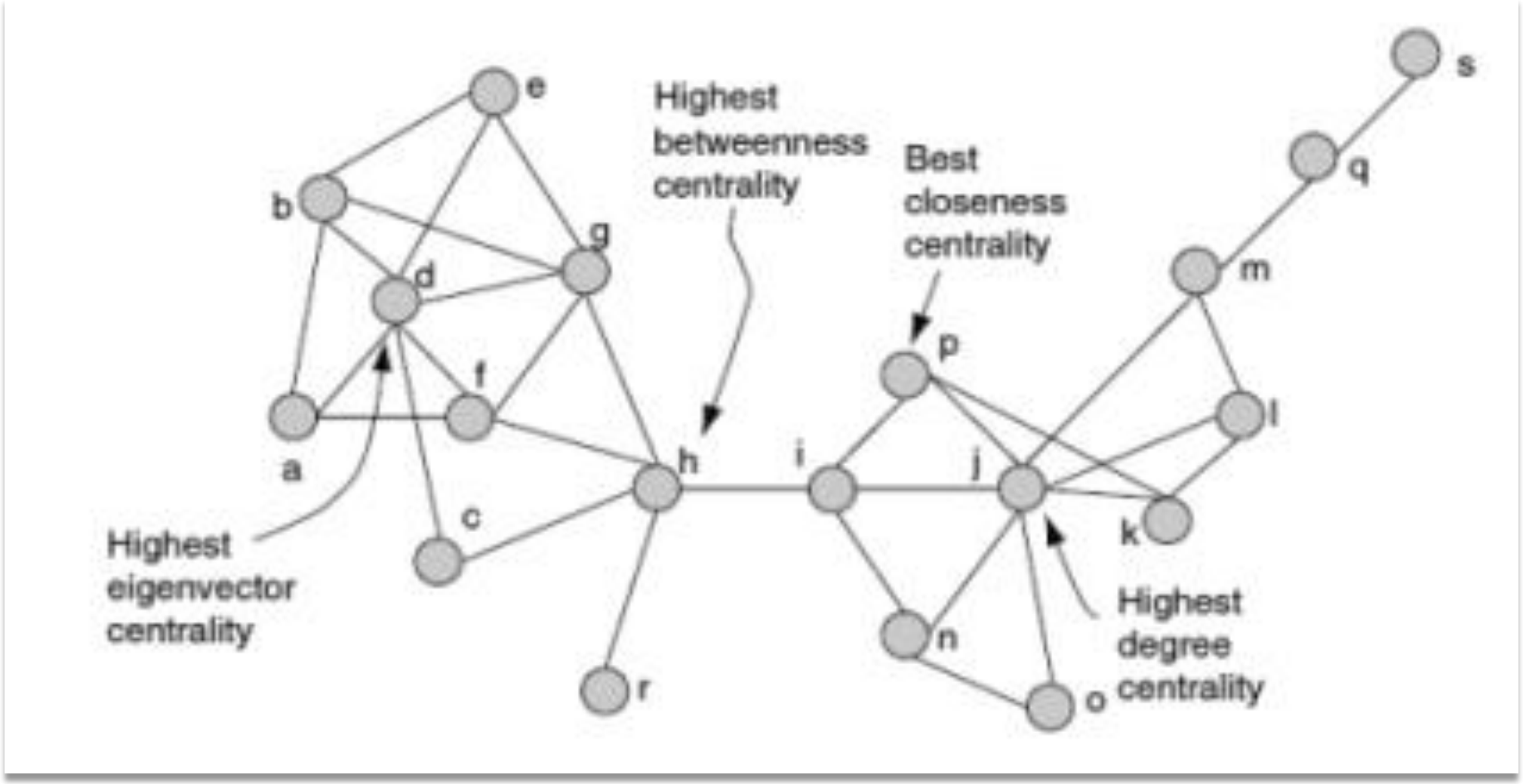


Image Source: Hendratno & Fitriati 2016. The study of Indonesia's readiness to cope with demographic bonus (DOI:10.22146/jieb.10311)

Eigenvector Centrality

For a given graph let $\boldsymbol{A} = (a_{i,j})$ be the adjacency matrix, where $a_{i,j}=1$ if vertex i is linked to vertex j, and $a_{i,j}=0$ otherwise.

The relative centrality score, x_i, of vertex i can be defined as:

$$x_i = \kappa_1^{-1} \sum_j A_{ij} x_j,$$

(Newman 2010, Eq. 7.6)

Where κ_i are the eigenvalues of $\boldsymbol{A}$, and κ_1 is the largest of them.

Eigenvector Centrality

Eigenvector centrality can be calculated for both undirected or directed networks, but it is more commonly used in undirected networks.

In a directed network the adjacency matrix is asymmetric, so it has two sets of eigenvectors, left and right eigenvectors, which allow you to measure how the node is influenced by others, and how it influences others.

You can think so them as analogous to **in-degree** and **out-degree** but more complex, as they capture the global structure of the network taking into account all nodes.

Eigenvector Centrality

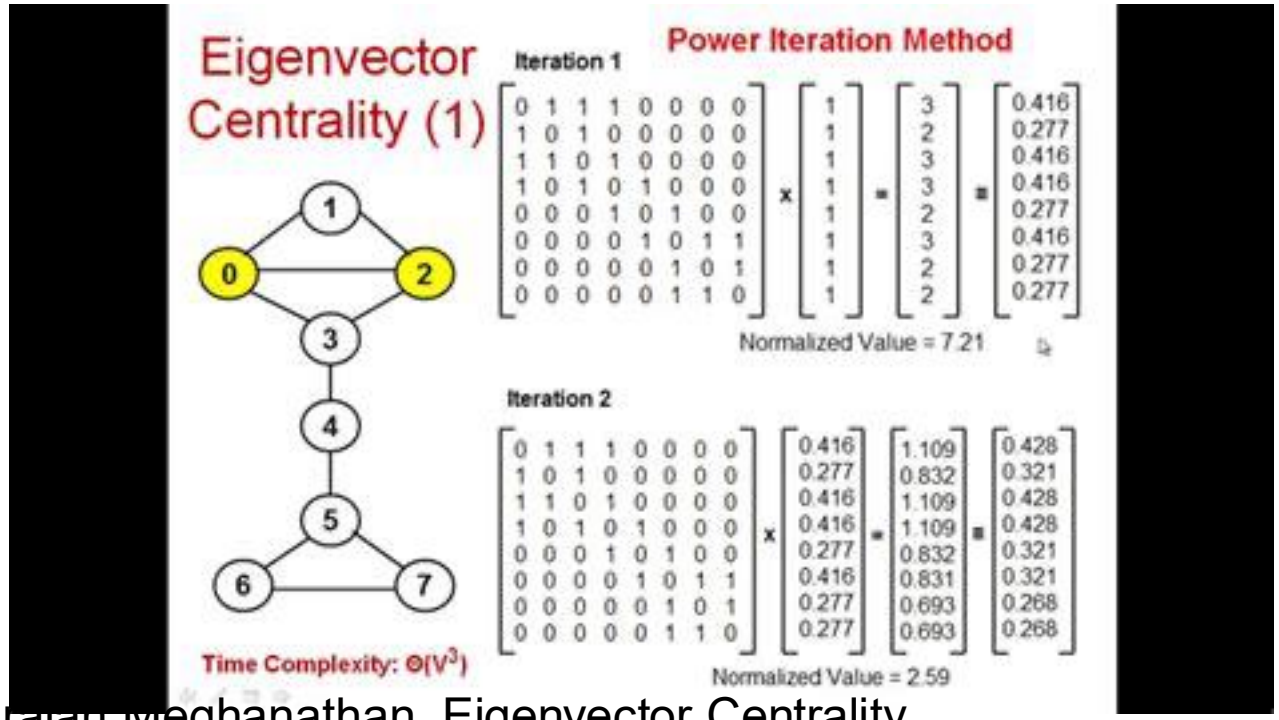
‘The centrality in directed networks is usually bestowed by other nodes pointing towards you, rather than by you pointing to others (so incoming links rather than outgoing links).

On the World Wide Web, for instance, the number and importance of web pages that point to your page can give a reasonable indication of how important or useful your page is.

The fact that your page might point to other important pages is irrelevant. Anyone can set up a page that points to a thousand others, but that does not make the page important.’

Source: Newman 2010, *Networks: An Introduction*

Eigenvector Centrality

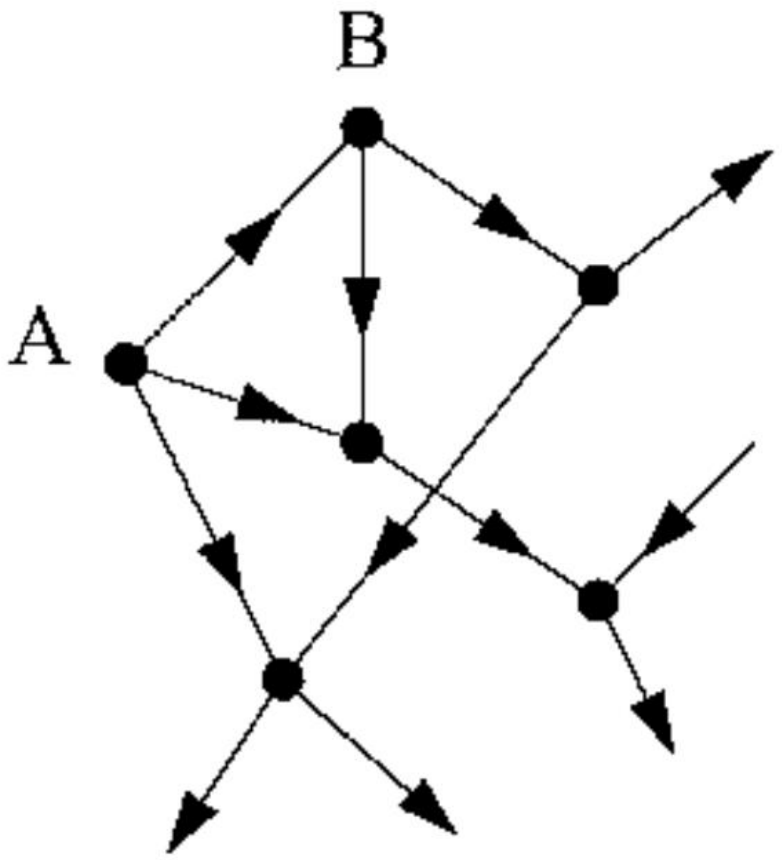


Natarajan Meghanathan, Eigenvector Centrality, https://www.youtube.com/watch?v=AjacGCIQ56o

Eigenvector Centrality

- In NetworkX, there are functions to compute the eigenvector centrality for the graph by adding the centrality of its predecessors. We will work with them in Lab5.
  - eigenvector_centrality_numpy(G, weight=None, max_iter=50, tol=0)
  - eigenvector_centrality(G, max_iter=100, tol=1e-06, nstart=None, weight=None)
- Source: https://networkx.org/documentation/stable/reference/algorithms/centrality.html

‘Vertex A in this figure is connected to the rest of the network but has only outgoing edges and no incoming ones.
Such a vertex will always have centrality zero because there are no terms in the sum defining eigenvector (Eq. 7.6).’
(Newman 2010, *Networks: An Introduction*)



Consider the equation and the position of vertex B in the graph.

Any Limitations?

$$x_i = \kappa_1^{-1} \sum_j A_{ij} x_j,$$

(Newman 2010, Eq. 7.6)

Katz Centrality

Katz Centrality

- A major problem with eigenvector centrality arises when it deals with directed graphs
- Centrality only passes over *outgoing* edges and in special cases such as when a node is in a directed acyclic graph centrality becomes zero
  - The node can have many edge connected to it
- To resolve this problem we add bias term β to the centrality values for all nodes

Elihu Katz

$$C_{\text{Katz}}(v_i) = \alpha \sum_{j=1}^{n} A_{j,i} C_{\text{Katz}}(v_j) + \beta$$

Activate Wind
Go to Settings to a

▶Shreesudha Kembhavi, EigenVector Centrality & Katz Centrality, https://www.youtube.com/watch?v=-LO9NLaccFQ

Katz Centrality

Katz Centrality - solution to the simple eigenvector centrality problem.
It gives each vertex a small amount of centrality 'for free' (*beta*) regardless of its position in the network (Eq. 7.8 In: Newman 2010, *Networks: An Introduction*)

$$x_i = \alpha \sum_j A_{ij} x_j + \beta,$$

(Newman 2010, Eq. 7.8)

Katz Centrality

- In NetworkX there are two functions to calculate katz centrality and we will test them in Lab5.

  katz_centrality_numpy(G, alpha=0.1, beta=1.0, normalized=True, weight=None)

  katz_centrality(G, alpha=0.1, beta=1.0, max_iter=1000, tol=1e-06, nstart=None, normalized=True, weight=None)

- Source: https://networkx.org/documentation/stable/reference/algorithms/centrality.html

Katz Centrality

Issue with Katz:

‘If a vertex with high Katz centrality points to many others then those others also get high centrality. A high-centrality vertex pointing to one million others gives all one million of them high centrality. One could argue-and many have-that this is not always appropriate’ (Newman 2010, *Networks: An Introduction*).

Solution:

Make it proportional to out-degree → **PageRank**

‘Then vertices that point to many others pass only a small amount of centrality on to each of those others, even if their own centrality is high’ (Newman 2010, *Networks: An Introduction*)

PageRank

‘PageRank works on the Web precisely because having links to your page from important pages elsewhere is a good indication that your page may be important too. But the added ingredient of dividing by the out-degrees of pages insures that pages that simply point to an enormous number of others do not pass much centrality on to any of them’. (Newman 2010, *Networks: An Introduction*)

The hubs do not have a disproportionate influence on the rankings of nodes around them.

PageRank

The formula for PageRank contains one the parameter *alpha,* whose value must be chosen before the algorithm can be used.

$$x_i = \alpha \sum_j A_{ij} \frac{x_j}{k_j^{\text{out}}} + \beta.$$

(Newman 2010, Eq. 7.15)

Its value should be less than the inverse of the largest eigenvalue (compare Eq. 7.6 and 7.8).

'The *Google* search engine uses a value of *alpha* = 0.85 in its calculations, although it's not clear that there is any rigorous theory behind this choice. More likely it is just a shrewd guess based on experimentation to find out what works well.' (Newman 2010, *Networks: An Introduction*)

Weight

Weight => Strength

Many real-world networks (directed and undirected) have link weights:
- Twitter/BlueSky diffusion / Email: number of messages / Air transportation: number of passengers

You can think of weights even in networks that we usually represent as unweighted:
- Facebook friendships could be weighted by number of interactions (tags, likes, comments, common friends)
- The movie co-star network could be weighted by number of movies in which two stars have acted together
- Wikipedia links can be weighted by number of clicks

In weighted networks degree, in-degree and out-degree turn into strength, in-strength, and out-strength

Weights and Influence

Weight is useful when, for example, working with Information Diffusion Networks.
To measure the influence of a an account in SoMe network we can use:

- Number of followers (in-degree in follower network)
- Number of users exposed (out-degree in retweet network)
- Fraction of retweets to followers
- **Number of retweets (out-strength in retweet network)**

Weights and Influence

eigenvector_centrality_numpy(G, **weight=None**, max_iter=50, tol=0)

eigenvector_centrality(G, max_iter=100, tol=1e-06, nstart=None, **weight=None**)

katz_centrality_numpy(G, alpha=0.1, beta=1.0, normalized=True, **weight=None**)

katz_centrality(G, alpha=0.1, beta=1.0, max_iter=1000, tol=1e-06, nstart=None, normalized=True, **weight=None**)

pagerank(G, alpha=0.85, personalization=None, max_iter=100, tol=1e-06, nstart=None, **weight='weight'**, dangling=None)

Still a bit lost?

At home,

- read this short blog post: Andrew Disney, ‘PageRank centra
EigenCentrality’, 2020,
https://cambridge-intelligence.com/eigencentrality-pagerank/

- watch this video demonstrating the difference between Eige
and PageRank using *Gephi:* Dr Alan Shaw, ‘An Overview of
Eigenvector Centrality and Pagerank for Social Networks’, 20
https://www.youtube.com/watch?v=kmQkXIg6Drw

Install *Gephi* before we meet next week!